310401

Roll No

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B. Tech. III - Sem. (Main) Exam., February - 2021

Computer Science & Engineering

3CS2-01 Advanced Engineering Mathematics

Common for CS/ IT

Time: 3 Hours **Maximum Marks: 120**

Instructions to Candidates:

Part – A: *Short answer questions (up to 25 words) 10 × 2 marks = 20 marks. All ten questions are compulsory.*

Part – B: *Analytical/Problem Solving questions (up to 100 words) 5 × 8 marks = 40 marks. Candidates have to answer five questions out of seven.*

Part – C: *Descriptive/Analytical/Problem Solving questions 4 × 15 marks = 60 marks. Candidates have to answer four questions out of five.*

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1. NIL

2. NIL

PART - A

Q.1 A continuous random variable X has the p.d.f. $f(x) = kx^2e^{-x}$, $x \geq 0$. Find k.

Q.2 Write the Pearson's β and γ coefficients based upon the first four moments about mean.

Q.3 If X is uniformly distributed with mean 1 and variance $\frac{4}{3}$. Find the parameters a and b for this distribution.

Q.4 Find the mean values for given lines of regression -

$$3x + 2y - 26 = 0,\ 6x + y - 31 = 0$$

Q.5 Define the design variables and design constraints.

https://www.btubikaner.com

Q.6 Define the feasible solution and optimal solution.

Q.7 Find the area of the largest rectangle whose perimeter be constant.

Q.8 Write the dual of the following -

$$\text{Min } z = 3x_1 + x_2$$

$$\text{s.t} \quad 2x_1 + 3x_2 \geq 2$$

$$x_1 + x_2 \geq 1$$

$$\text{and} \quad x_1, x_2 \geq 0$$

Q.9 Find the minimum total unit cost of the following assignment problem -

		Plant		
		1	2	3
Product	1	10	8	12
	2	18	6	14
	3	6	4	2

Q.10 What do you mean by slack and surplus variables in simplex method?

PART – B

Q.1 Let X be a random variable with the following probability distribution -

x:	-3	6	9
P(X = x):	1/6	1/2	1/3

Find $E(X)$, $E(X^2)$, $E(2X+1)^2$

Q.2 A fair die is tossed 720 times. Use Chebyshev's inequality to find a lower bound for the probability of getting 100 to 140 sixes. https://www.btubikaner.com

Q.3 Fit a parabola of second degree to the following data -

X:	0	1	2	3	4
Y:	1	1.8	1.3	2.5	6.3

Q.4 Find the Moment generating function of Poisson and Exponential distribution.

Q.5 Write the applications of optimization in engineering field.

Q.6 Use Lagrangian theory to find the volume of the greatest parallelepiped inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{x^2}{c^2} = 1$.

Q.7 Solve the following LPP by Simplex method -

Min. $z = x_1 + x_2$

s.t. $2x_1 + x_2 \geq 4$

$x_1 + 7x_2 \geq 7$

and $x_1, x_2 \geq 0$

PART – C

Q.1 The joint density function of continuous random variables X and Y is -

$$f(x, y) = \begin{cases} \frac{1}{3}(x + y), & 0 < x < 1, \quad 0 < y < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Find the Var(X), Var(Y) and correlation coefficient between X and Y.

Q.2 In a Normal distribution, 7% of the items are under 35 and 89% are under 63. What are the mean and standard deviation of the distribution? It is given that -

$$f(t) = \frac{1}{\sqrt{2\pi}} \int_0^t \exp\left(\frac{-x^2}{2}\right) dx$$

then $f(1.23) = 0.39$, $f(1.48) = 0.43$

Q.3 Explain the classification of optimization problems.

Q.4 Use Kuhn-Tucker conditions, determine x_1, x_2 and x_3 so as to -

Max. $z = -x_1^2 - x_2^2 - x_3^2 + 4x_1 + 6x_2$

s.t $x_1 + x_2 \leq 2$

$2x_1 + 3x_2 \leq 12$

and $x_1, x_2 \geq 0$

Q.5 Find the optimal solution of the following Transportation problem -

	W_1	W_2	W_3	W_4	Capacity ↓
F_1	19	30	50	10	7
F_2	70	30	40	60	9
F_3	40	8	70	20	18
Demand →	5	8	7	14	34
